CC Lab: 6

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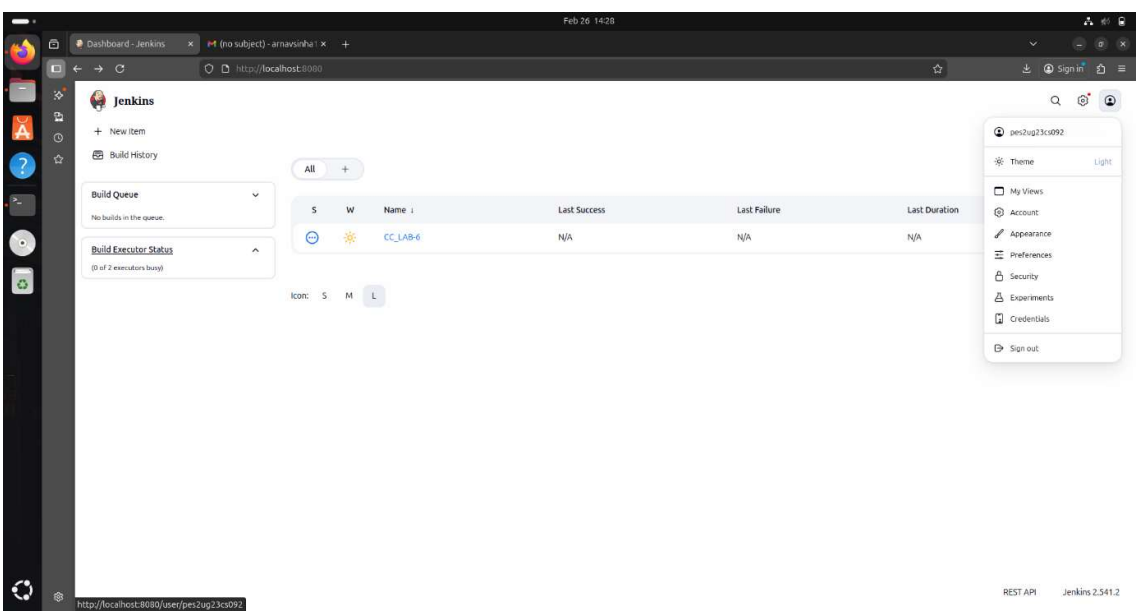
Github: https://github.com/arnavsinha20/CC_LAB-6

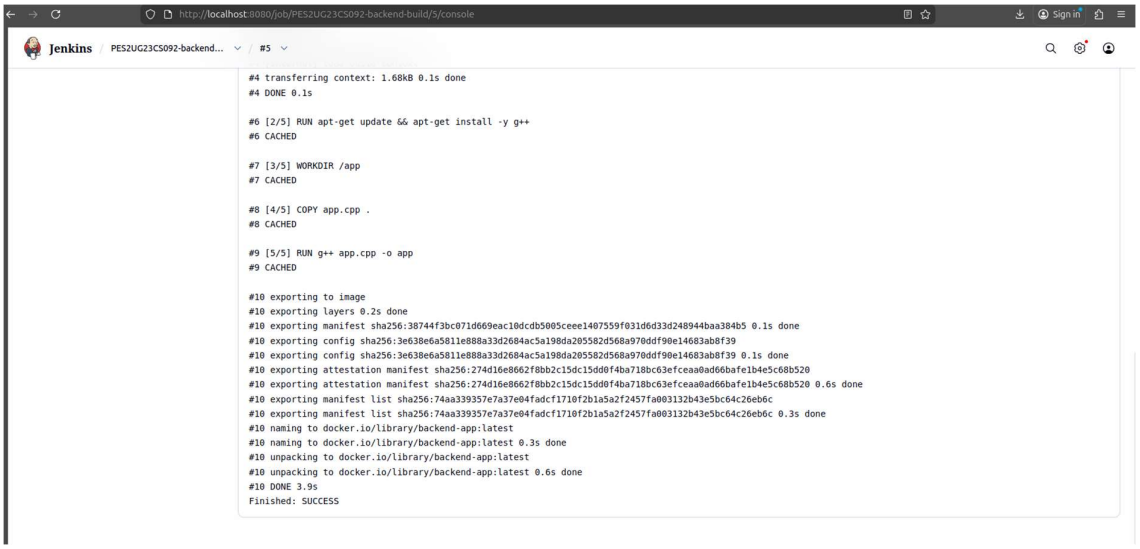
SS1

```
pes2ug23cs092@pes2ug23cs092:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED       STATUS       PORTS
NAMES
c9fc5312ec6f   jenkins/jenkins:lts    "/usr/bin/tini -- /u…"   2 hours ago   Up 2 hours   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
jenkins
pes2ug23cs092@pes2ug23cs092:~$ docker logs jenkins
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-02-26 06:31:40.313+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-02-26 06:31:47.017+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-02-26 06:31:47.129+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; j
vm 21.0.9+10-LTS
2026-02-26 06:31:49.180+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-02-26 06:31:49.619+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-02-26 06:31:50.872+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-02-26 06:31:51.261+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3681037{Jenkins v2.541.2,/,b=file:///var/jenkin
s_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@2459319c{STARTED}}
2026-02-26 06:31:51.312+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@2aa6311a{HTTP/1.1, (http/1.1){0.0.0.0:8080}
2026-02-26 06:31:51.316+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@53a665ad{STARTING}[12.1.5,sto=0] @4191ms
2026-02-26 06:31:51.341+0000 [id=41]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-02-26 06:31:51.550+0000 [id=40]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.541.2
2026-02-26 06:31:51.736+0000 [id=50]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-02-26 06:31:51.939+0000 [id=66]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-02-26 06:31:54.653+0000 [id=66]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-02-26 06:31:54.667+0000 [id=51]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-02-26 06:31:54.671+0000 [id=51]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-02-26 06:31:56.008+0000 [id=65]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-02-26 06:31:56.029+0000 [id=65]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-02-26 06:31:56.042+0000 [id=65]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-02-26 06:31:56.085+0000 [id=49]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-02-26 06:31:56.415+0000 [id=80]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-02-26 06:32:00.034+0000 [id=58]    INFO    jenkins.install.SetupWizard#init:
[LF]>
[LF]> *************************************************************
[LF]> *************************************************************
[LF]> *************************************************************
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 99cf3a334b9843159609589f469a2448
```

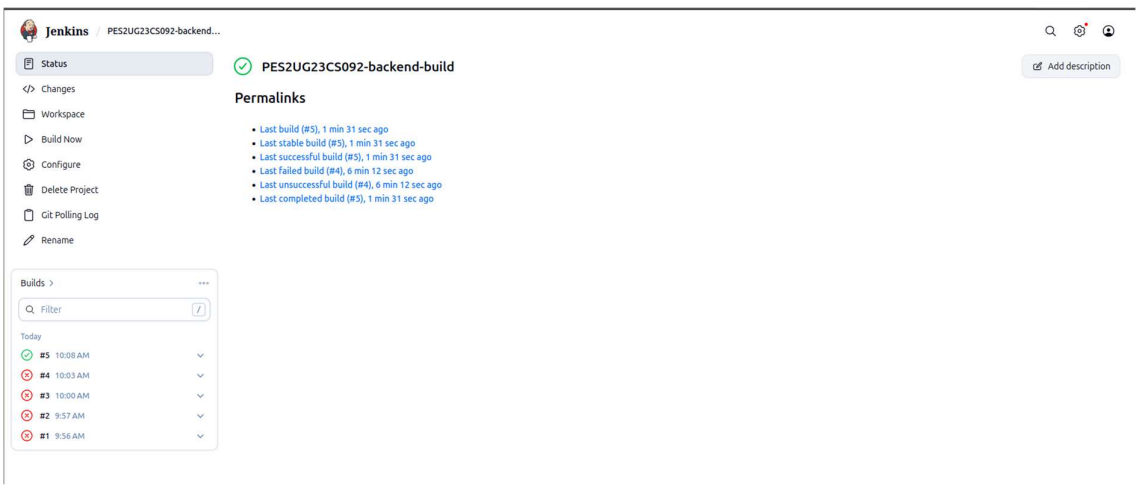
SS2



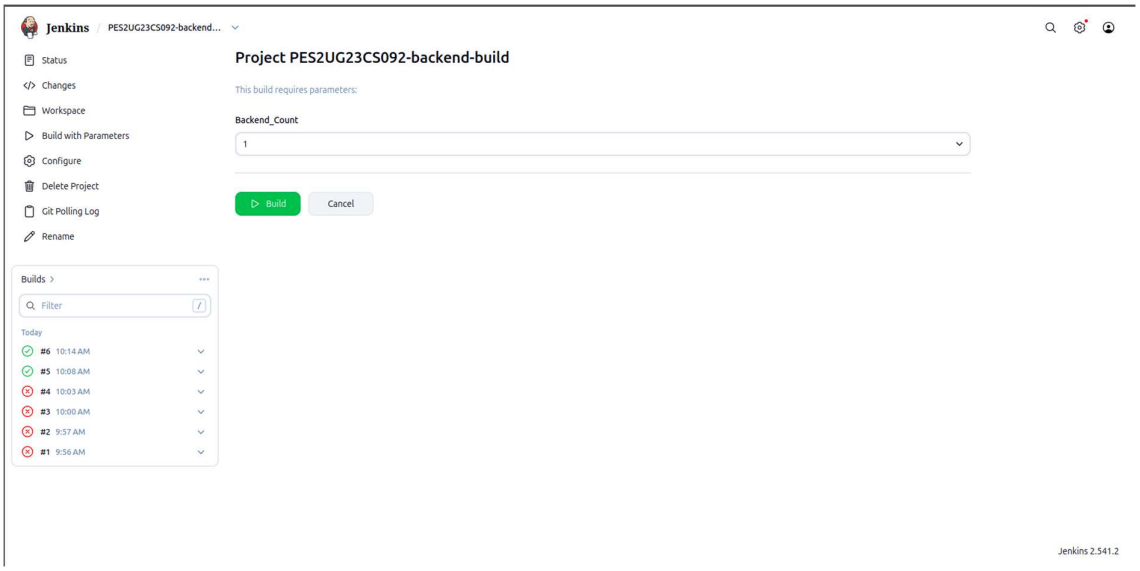
SS3:

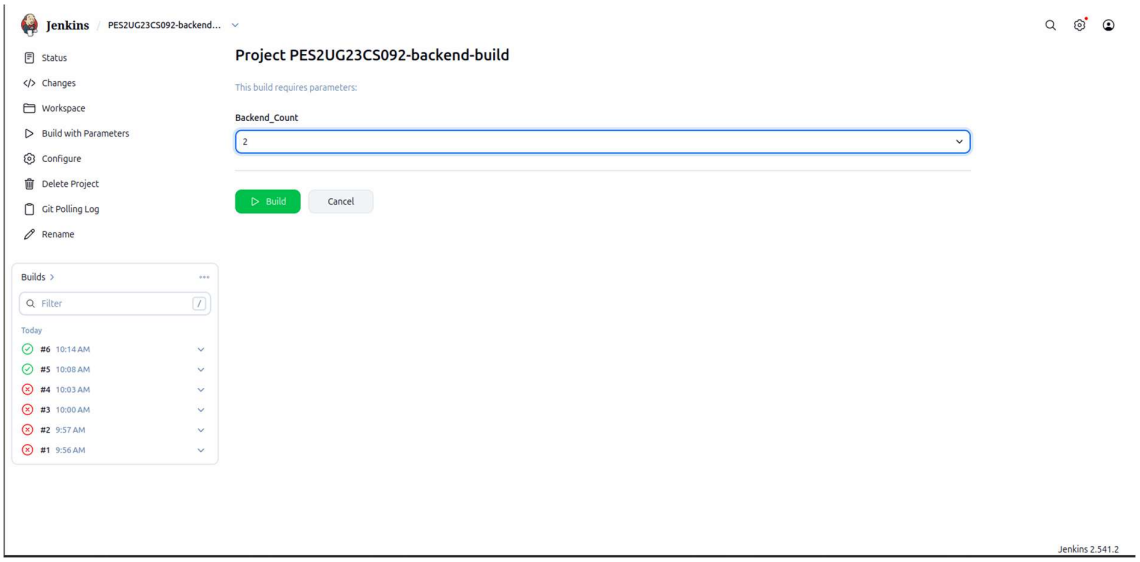

http://localhost:8080/job/PES2UG23CS092-backend-build/5/console
Sign in
Jenkins
PES2UG23CS092-backend...
#5
#4 transferring context: 1.68kB 0.1s done
#4 DONE 0.1s
#6 [2/5] RUN apt-get update && apt-get install -y g++
#6 CACHED
#7 [3/5] WORKDIR /app
#7 CACHED
#8 [4/5] COPY app.cpp .
#8 CACHED
#9 [5/5] RUN g++ app.cpp -o app
#9 CACHED
#10 exporting to image
#10 exporting layers 0.2s done
#10 exporting manifest sha256:38744f3bc071d669eac10dcdb5005ceee1407559f031d6d33d248944baa384b5 0.1s done
#10 exporting config sha256:3e638e6a5811e888a33d2684ac5a198da205582d568a970ddf90e14683ab8f39
#10 exporting config sha256:3e638e6a5811e888a33d2684ac5a198da205582d568a970ddf90e14683ab8f39 0.1s done
#10 exporting attestation manifest sha256:274d16e8662f8bb2c15dc15dd0f4ba718bc63efceaa0ad66bafe1b4e5c68b520
#10 exporting attestation manifest sha256:274d16e8662f8bb2c15dc15dd0f4ba718bc63efceaa0ad66bafe1b4e5c68b520 0.6s done
#10 exporting manifest list sha256:74aa339357e7a37e04fadcf1710f2b1a5a2f2457fa003132b43e5bc64c26eb6c
#10 exporting manifest list sha256:74aa339357e7a37e04fadcf1710f2b1a5a2f2457fa003132b43e5bc64c26eb6c 0.3s done
#10 naming to docker.io/library/backend-app:latest
#10 naming to docker.io/library/backend-app:latest 0.3s done
#10 unpacking to docker.io/library/backend-app:latest
#10 unpacking to docker.io/library/backend-app:latest 0.6s done
#10 DONE 3.9s
Finished: SUCCESS

SS4:

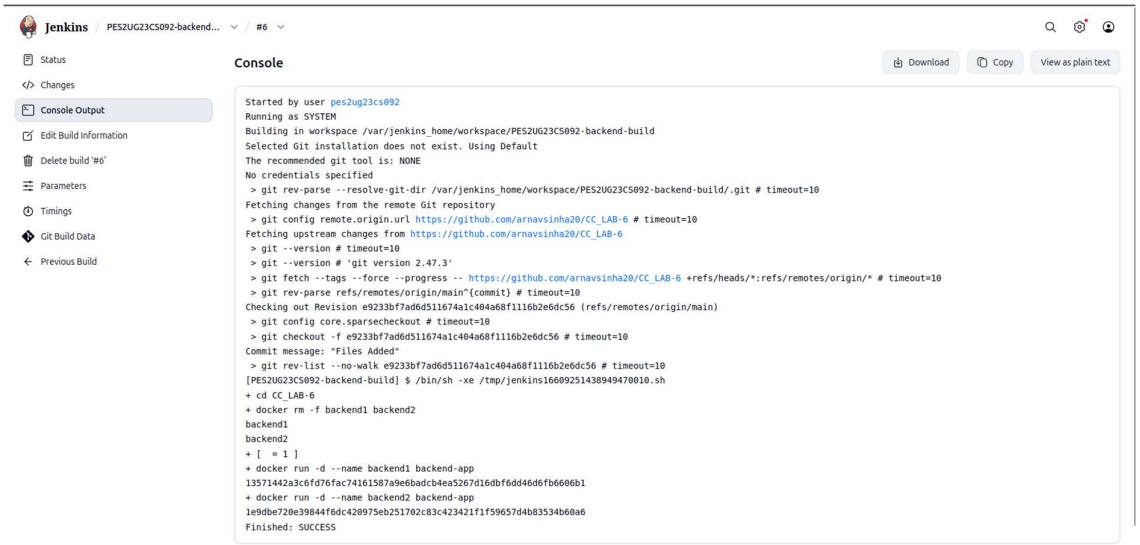

Jenkins
PES2UG23CS092-backend...
Status
Changes
Workspace
Build Now
Configure
Delete Project
Git Polling Log
Rename
Builds
Filter
Today
#5 10:08 AM
#4 10:03 AM
#3 10:00 AM
#2 9:57 AM
#1 9:56 AM
PES2UG23CS092-backend-build
Add description
Permalinks
Last build (#5), 1 min 31 sec ago
Last stable build (#5), 1 min 31 sec ago
Last successful build (#5), 1 min 31 sec ago
Last failed build (#4), 6 min 12 sec ago
Last unsuccessful build (#4), 6 min 12 sec ago
Last completed build (#5), 1 min 31 sec ago

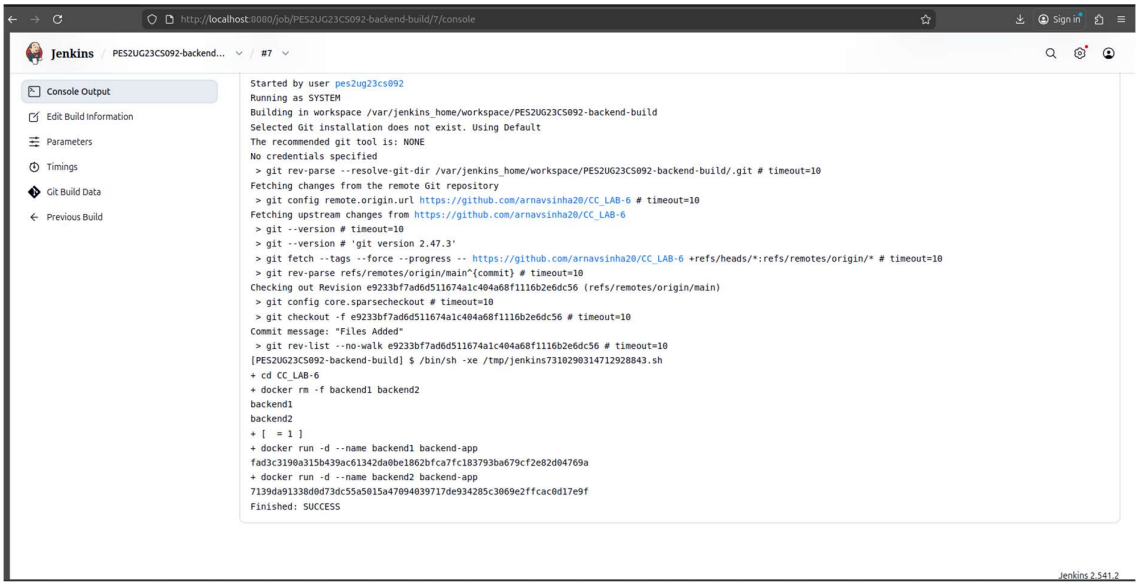
SS5:


Jenkins
PES2UG23CS092-backend...
Status
Changes
Workspace
Build with Parameters
Configure
Delete Project
Git Polling Log
Rename
Project PES2UG23CS092-backend-build
This build requires parameters:
Backend_Count
1
Build
Cancel
Builds
Filter
Today
#6 10:14 AM
#5 10:08 AM
#4 10:03 AM
#3 10:00 AM
#2 9:57 AM
#1 9:56 AM
Jenkins 2.541.2

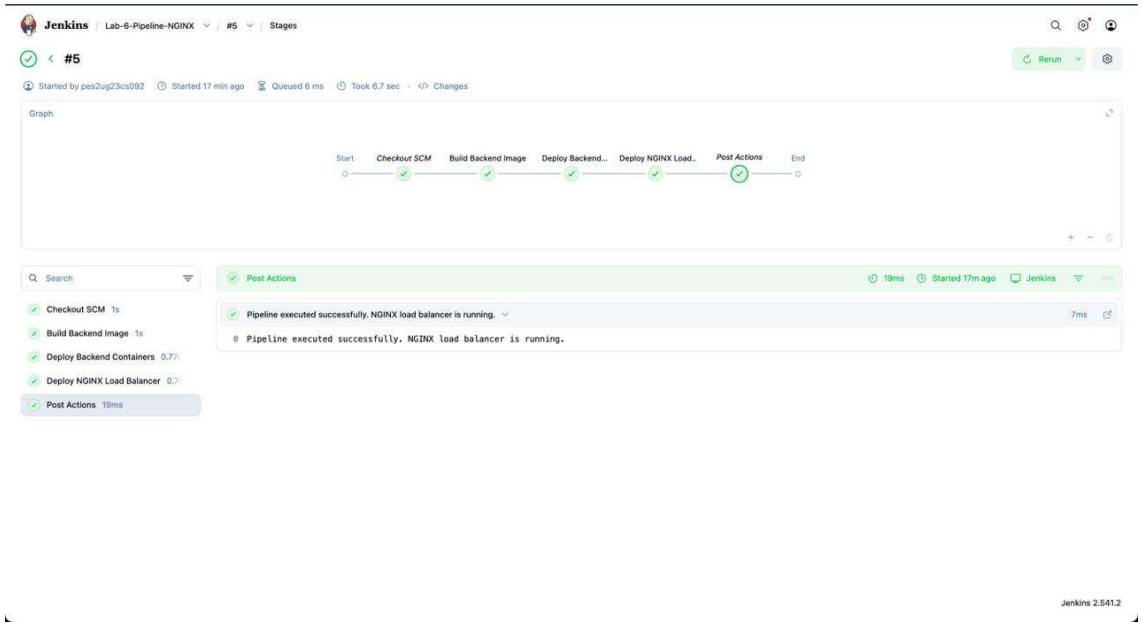

Jenkins
PES2UG23CS092-backend...
Status
Changes
Workspace
Build with Parameters
Configure
Delete Project
Git Polling Log
Rename
Project PES2UG23CS092-backend-build
This build requires parameters:
Backend_Count
2
Build
Cancel
Builds
Filter
Today
#6 10:14 AM
#5 10:08 AM
#4 10:03 AM
#3 10:00 AM
#2 9:57 AM
#1 9:56 AM
Jenkins 2.541.2

SS6:

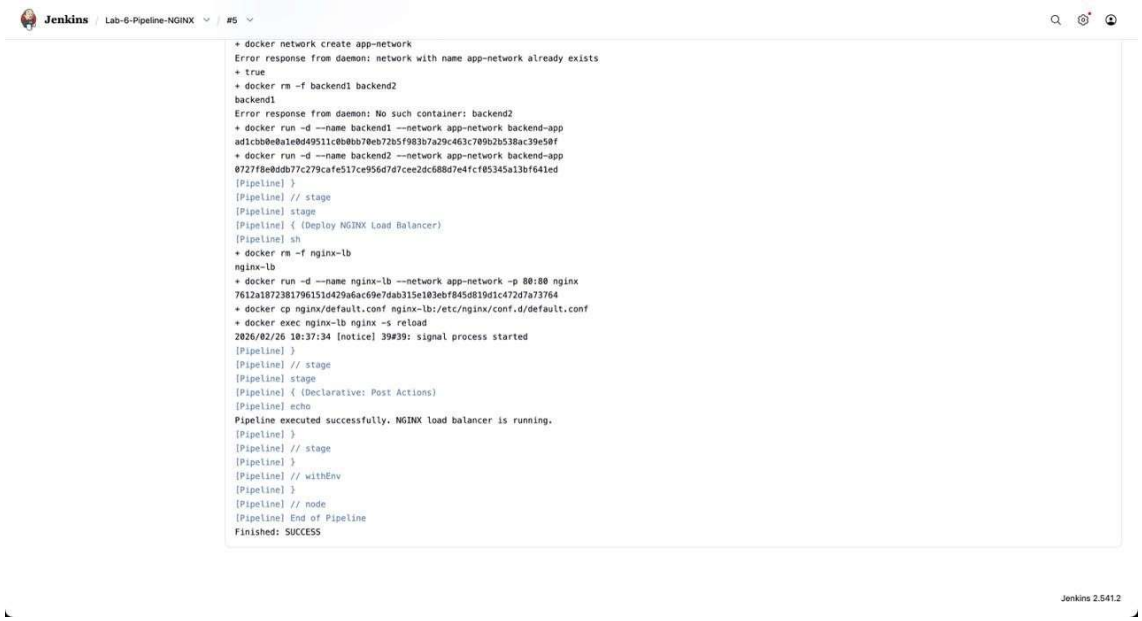

Jenkins
PES2UG23CS092-backend...
#6
Status
Changes
Console Output
Edit Build Information
Delete build '#6'
Parameters
Timings
Git Build Data
Previous Build
Console
Download
Copy
View as plain text
Started by user pes2ug23cs092
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/PES2UG23CS092-backend-build
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/PES2UG23CS092-backend-build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/arnavsinha20/CC_LAB-6 # timeout=10
Fetching upstream changes from https://github.com/arnavsinha20/CC_LAB-6
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/arnavsinha20/CC_LAB-6 +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision e9233bf7ad6d511674a1c404a68f1116b2e6dc56 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f e9233bf7ad6d511674a1c404a68f1116b2e6dc56 # timeout=10
Commit message: "Files Added"
> git rev-list --no-walk e9233bf7ad6d511674a1c404a68f1116b2e6dc56 # timeout=10
[PES2UG23CS092-backend-build] $ /bin/sh -xe /tmp/jenkins16609251438949470010.sh
+ cd CC_LAB-6
+ docker rm -f backend1 backend2
backend1
backend2
+ [= 1]
+ docker run -d --name backend1 backend-app
13571442a3c6fd76fac74161587a9e6badcb4ea5267d16dbf6dd46d6fb6606b1
+ docker run -d --name backend2 backend-app
1e9dbe720e39844f6dc420975eb251702c83c423421f1f59657d4b83534b60a6
Finished: SUCCESS


http://localhost:8080/job/PES2UG23CS092-backend-build/7/console
Sign in
Jenkins
PES2UG23CS092-backend...
#7
Console Output
Edit Build Information
Parameters
Timings
Git Build Data
Previous Build
Started by user pes2ug23cs092
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/PES2UG23CS092-backend-build
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/PES2UG23CS092-backend-build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/arnavsinha20/CC_LAB-6 # timeout=10
Fetching upstream changes from https://github.com/arnavsinha20/CC_LAB-6
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/arnavsinha20/CC_LAB-6 +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision e9233bf7ad6d511674a1c404a68f1116b2e6dc56 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f e9233bf7ad6d511674a1c404a68f1116b2e6dc56 # timeout=10
Commit message: "Files Added"
> git rev-list --no-walk e9233bf7ad6d511674a1c404a68f1116b2e6dc56 # timeout=10
[PES2UG23CS092-backend-build] $ /bin/sh -xe /tmp/jenkins7310290314712928843.sh
+ cd CC_LAB-6
+ docker rm -f backend1 backend2
backend1
backend2
+ [= 1]
+ docker run -d --name backend1 backend-app
fad3c3190a315b439ac61342da0be1862bfca7fc183793ba679cf2e82d04769a
+ docker run -d --name backend2 backend-app
7139da91338d0d73dc55a5015a47094039717de934285c3069e2ffcac0d17e9f
Finished: SUCCESS
Jenkins 2.541.2

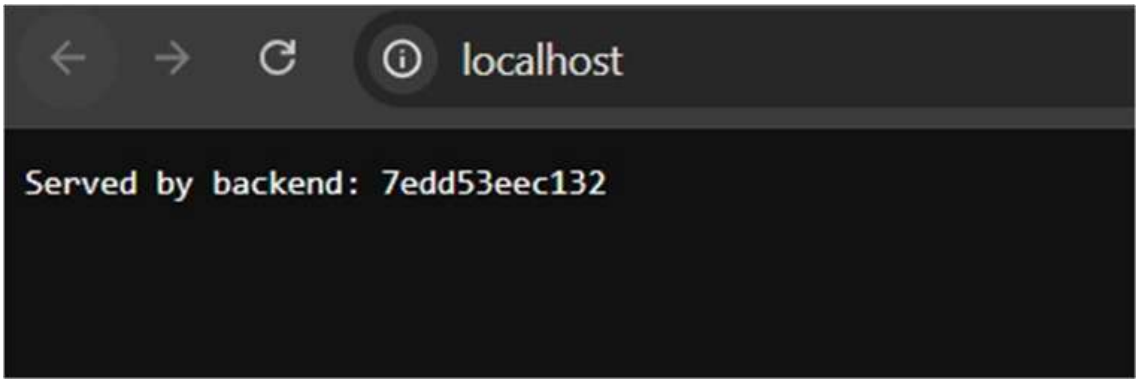
SS7:

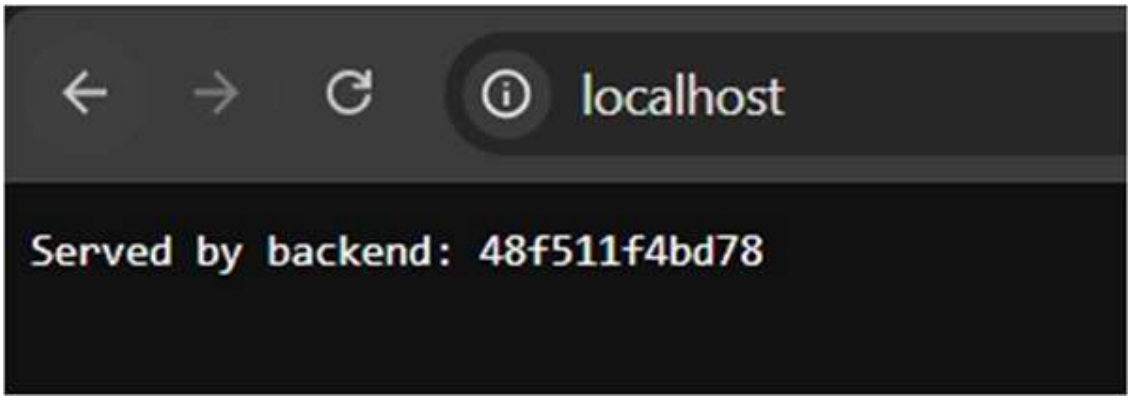

Jenkins / Lab-6-Pipeline-NGINX / #5 / Stages
#5
Rerun
Started by pes2ug23cs092
Started 17 min ago
Queued 6 ms
Took 6.7 sec
Changes
Graph
Start
Checkout SCM
Build Backend Image
Deploy Backend...
Deploy NGINX Load...
Post Actions
End
Search
Checkout SCM 1s
Build Backend Image 1s
Deploy Backend Containers 0.7
Deploy NGINX Load Balancer 0.7
Post Actions 19ms
Post Actions
19ms
Started 17m ago
Jenkins
Pipeline executed successfully. NGINX load balancer is running.
7ms
0 Pipeline executed successfully. NGINX load balancer is running.
Jenkins 2.541.2

SS8:

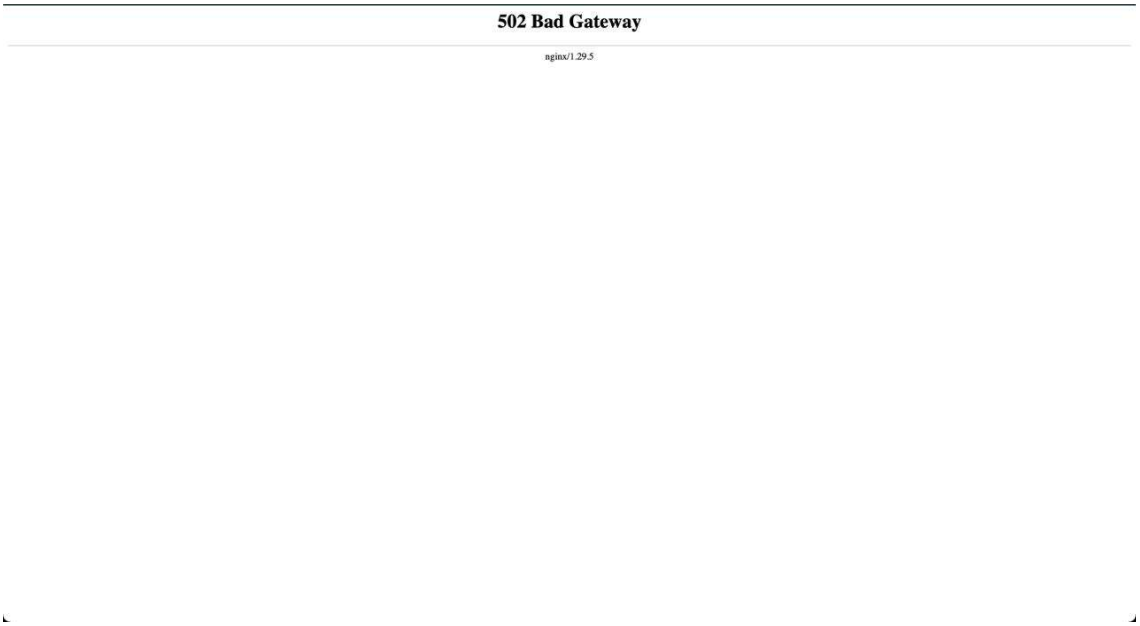

Jenkins / Lab-6-Pipeline-NGINX / #5
+ docker network create app-network
Error response from daemon: network with name app-network already exists
+ true
+ docker rm -f backend1 backend2
backend1
Error response from daemon: No such container: backend2
+ docker run -d --name backend1 --network app-network backend-app
ad1cbb0e0a1e0d49511c0b0bb70eb72b5f983b7a29c463c709b2b538ac39e50f
+ docker run -d --name backend2 --network app-network backend-app
0727f8e0ddb77c279cafe517ce956d7d7cee2dc688d7e4fcf05345a13bf641ed
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy NGINX Load Balancer)
[Pipeline] sh
+ docker rm -f nginx-lb
nginx-lb
+ docker run -d --name nginx-lb --network app-network -p 80:80 nginx
7612a187238179615 1d429a6ac69e7dab315e103ebf845d819d1c472d7a73764
+ docker cp nginx/default.conf nginx-lb:/etc/nginx/conf.d/default.conf
+ docker exec nginx-lb nginx -s reload
2026/02/26 10:37:34 [notice] 39#39: signal process started
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully. NGINX load balancer is running.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.2

SS9:


localhost
Served by backend: 7edd53eec132


localhost
Served by backend: 48f511f4bd78

SS10:


502 Bad Gateway
nginx/1.29.5